

FLYROTOR ESC Compatibility with S.BUS2 Protocol Instruction Manual

v1.2 (20260608)

1. Sensor Types

Curr.F1678

- ① **Voltage:** Battery voltage [0–81.91V]
- ② **Current:** Battery current [0–163.83A]
- ③ **Capacity:** Consumed capacity [0–32766mAh]

Voltage Sensor

- ① **Battery:** SBEC voltage [0–50V]
- ② **External battery:** Battery current [0–819.1A]

RPM Sensor

- ① **RPM:** Main rotor head speed [0–393210 RPM]

Temperature Sensor

- ① **Temperature:** MOSFET temperature value [0–150°C]

Temp-F1713

- ① **Temperature:** Motor temperature value [0–150°C]

2. Usage Instructions

① **Hardware connection:** Connect the purple wire of the ESC's T-port to the S.BUS2 port of a telemetry-supported receiver.

② **ESC setting:** Set the ESC telemetry protocol to S.BUS2.

③ **Adding sensors on the transmitter:**

Procedure: Home screen → Linkage Menu → Sensors

Add the following sensor types:

Data Slot 1: Curr.F1678

Data Slot 4: Voltage Sensor

Data Slot 6: RPM Sensor

Data Slot 8: Temperature Sensor

Data Slot 9: Temp-F1713

④ **RPM sensor setup:**

Procedure: Home screen → Linkage Menu → Telemetry → RPM Sensor

Sensor type: Magnetic

Gear ratio calculation formula:

$\text{Gear ratio} = (\text{Main gear teeth} / \text{Pinion teeth}) \times (\text{Motor poles} \div 2)$

Example: Main gear 107T, pinion 12T, motor poles 10

$\text{Gear ratio} = (107 \div 12) \times (10 \div 2) = 44.58$

3. Devices Supporting S.BUS2 Telemetry

Receivers:

T-FHSS: R3001SB, R3004SB, R3006SB, R3008SB

FASSTest: R7003SB, R7103SB, R7006SB, R7106SB, R7008SB, R7108SB, R7208SB, R7308SB

Gyro: CGY760R

Transmitters: T16SZ, T16IZ, T18SZ, T18MZ, T32MZ

Note: The above list of devices is not exhaustive.